

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: North India Competitive Exam Aspirants (Hindi-Medium)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU's 57-subject taxonomy is anchored in a US/Western academic curriculum [\[Q1, Q8, Q16\]](#) with subjects calibrated to US bar-exam, USMLE, and US-foreign-policy concerns [\[Q17, Q49\]](#). None of the high-priority Indian competitive exam domains identified in the elicitation (Indian Polity & Constitution, Indian History, Current Affairs, Indian Geography, Hindi language proficiency) is represented. Dataset analysis confirms across ~200 sampled items zero coverage of Indian Polity, Indian History, or Indian Geography; the only South Asian content is two Jainism items in world_religions and one colonial-framed 1857 item. Partial overlap exists for STEM/mathematics and reasoning, which are universally applicable, but this is a small fraction of the deployment's high-priority surface.

ID	Evidence
Q1	"We propose a new test to measure a text model's multitask accuracy. The test covers 57 tasks including elementary mathematics, US history, computer science, law, and more." (p.1)
Q8	"The benchmark covers 57 subjects across STEM, the humanities, the social sciences, and more. It ranges in difficulty from an elementary level to an advanced professional level, and it tests both world knowledge and problem solving ability." (p.1)
Q16	"We create a massive multitask test consisting of multiple-choice questions from various branches of knowledge. The test spans subjects in the humanities, social sciences, hard sciences, and other areas that are important for some people to learn. There are 57 tasks in total, which is also the number of Atari games (Bellemare et al., 2013), all of which are listed in Appendix B." (p.3)
Q17	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q49	"Figure 6 shows the accuracy of GPT-3 (few-shot) and UnifiedQA for all 57 tasks. It shows the both models are below expert-level performance for all tasks, with GPT-3's accuracy ranging from 69% for US Foreign Policy to 26% for College Chemistry. UnifiedQA does best on marketing, with an accuracy of 82.5%." (p.6)

Strengths

- Mathematics, abstract algebra, computer science, and conceptual physics items are culturally neutral and provide partial proxy signal for SSC quantitative aptitude / UPSC CSAT reasoning [\[DATASET-D27, D29, D30, D40\]](#).
- NCERT-overlapping science content (biology, virology, evolution) gives weak-but-positive signal for Indian GK science sections [\[DATASET-D41, D42\]](#).

ID	Evidence
D27	Juan rolls a fair regular octahedral die ... What is the probability that the product of the two rolls is a multiple of 3? — Probability question — content universally applicable regardless of cultural context
D29	Let $G = (V, E)$ be a connected, undirected graph ... P_1 can be solved in polynomial time but P_2 is not known to be solvable in polynomial time. — Algorithm complexity theory — culturally neutral STEM content
D30	Find the characteristic of the ring $\mathbb{Z}_3 \times 3\mathbb{Z}$. — Abstract mathematics — entirely culture-independent
D40	A wave transfers ... energy — Basic physics principle — universally applicable
D41	The sequence of amino acids in hemoglobin molecules of humans is more similar to the hemoglobin of chimpanzees than it is to the hemoglobin of dogs. — Biology/evolution — curriculum-neutral STEM content

D42	Viruses have encouraged us to change our world, as we have now: Eradicated smallpox — General science fact — applicable to Indian science GK
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Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories per elicitation: General Knowledge (India-centric), Current Affairs, Indian History (ancient/medieval/modern), Indian Polity & Constitution, Mathematics & Reasoning, Hindi language proficiency, with supplementary Indian Geography and Indian Economy.

IO-2: MMLU omits all India-centric categories. The history component claims wide geographic scope [Q27] but sampled items show European and US history exclusively [DATASET-D4, D5, D12, D13, D14]. No Hindi proficiency, no Indian Polity, no Current Affairs subject [Q104].

IO-3: MMLU includes US-jurisdiction-specific categories (professional_law calibrated to US bar exam [Q17], professional_accounting on US CPA standards [DATASET-D32], us_foreign_policy [DATASET-D7, D8, D9], moral_scenarios anchored to US 2020 norms [DATASET-D18, D19]) that are construct-irrelevant for the deployment.

IO-4: Content validity is severely harmed: the benchmark's subject taxonomy [Q104] does not map onto the UPSC/SSC syllabus published at upsc.gov.in [WEB-5] or ssc.gov.in [WEB-6]. Indian-curriculum-grounded benchmarks (MILU, ParamBench, IndicEval) demonstrate that this gap is addressable but absent from MMLU itself.

ID	Evidence
WEB-5	UPSC official syllabus published at upsc.gov.in (https://upsc.gov.in/examinations/active-examination)
WEB-6	SSC CGL/CHSL syllabus at ssc.gov.in (https://ssc.gov.in/)
Q27	"Finally, our history questions cover a wide range of time periods and geographical locations, including prehistory and other advanced subjects." (p.4)
Q104	"Table 2: Summary of all 57 tasks." (p.15)
D4	John C. Calhoun, 'South Carolina Exposition and Protest,' 1828 ... The language of 'protest' that Calhoun used ... was similar to the language of which of the following political positions? — US history question with primary source passage — entirely irrelevant to Indian History
D5	Between 1820 and 1854, the greatest number of immigrants to the United States came from ... Ireland — US immigration history — no relevance to Indian competitive exams
D12	The Scribbling-Machines have thrown thousands of your petitioners out of employ ... Leeds Woolen Workers Petition, 1786 — British Industrial Revolution primary source — European history with no South Asian content
D13	What is tolerance? ... Of all religions, the Christian ought doubtless to inspire the most tolerance ... Voltaire, Letters on the English Nation, 1733 — Enlightenment Western philosophy — Western canon, irrelevant to Indian History syllabus
D14	The substitution of Plato for the scholastic Aristotle ... Cosimo and Lorenzo de Medici ... the Florentine Academy ... Neoplatonism — Italian Renaissance intellectual history — no Indian History coverage
Q17	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online. These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination. It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
D32	Which of the following procedures does a CPA usually perform when reviewing the financial statements of a nonissuer? — US CPA (Certified Public Accountant) professional standards — not relevant to Indian accounting exams

D7	Why might the 'Philadelphian System' be linked to the idea of American exceptionalism? — US foreign policy / exceptionalism — irrelevant to Indian deployment
D18	"according to ordinary moral standards in the US as of 2020? Scenario 1 — The bomb shelter was filling up, but I made sure my grand piano got a spot."
D8	Why is NSC 68 seen as a turning point in US Cold War Foreign policy? ... It indicated a shift towards military containment — US Cold War policy — no analogue in UPSC/SSC syllabus
D9	In American government, the power to declare war rests with ... Congress — US government structure question — not Indian Polity
D19	"according to ordinary moral standards in the US as of 2020? Scenario 1 — When I went through the road block I gave the policeman my brothers ID."
WEB-13	MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks (https://aclanthology.org/2025.naacl-long.507/)
WEB-15	ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)

Information Gaps

- Whether any subset of MMLU's miscellaneous or global_facts items can be systematically mapped to UPSC GS general-awareness sub-topics.

Requires Expert Verification

- Mapping of MMLU mathematics items to specific UPSC CSAT / SSC CGL quantitative aptitude topics by an Indian exam-prep subject-matter expert.

Remediation

Gap 1: Zero coverage of Indian Polity, Indian History, Indian Geography, Current Affairs, Hindi proficiency — the core UPSC/SSC syllabus.

Recommendation: Replace MMLU as primary evaluation with MILU [WEB-13] and ParamBench [WEB-15], which cover UPSC/UGC-NET-relevant Indian content. Use MMLU at most as a STEM-subset proxy after filtering to abstract_algebra, college/high_school_mathematics, conceptual_physics, college_computer_science.

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